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STUDY OF INDUSTRIAL GROWTH AND PERFORMANCE

An Empirical Analysis of India's Manufacturing Sector, 1981-82 to 2023-24

Based on the Annual Survey of Industries (ASI) 2023-24, Ministry of Statistics and Programme Implementation (MoSPI)

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Executive Summary

This report presents a comprehensive statistical and machine-learning analysis of India's organized manufacturing sector, drawing on four official data products of the Annual Survey of Industries (ASI) 2023-24 published by MoSPI: the 43-year national annual time series (1981-82 to 2023-24), the 29-industry cross-section, the 36-state/UT cross-section, and the 77-industry granular NIC 3-digit breakdown. Across every table used, the sum of constituent units reconciles with the published All-India control total to within rounding error, confirming the underlying data is internally consistent.

Three findings anchor the report. First, India's industrial output, net value added (NVA), and fixed capital have grown at compound annual rates of 10.51%, 9.72%, and 12.35% respectively since 1981-82 (nominal), while the workforce grew at only 2.25% per year — industrial growth over four decades has been overwhelmingly capital-driven, not employment-driven. Second, production is geographically and sectorally concentrated but not to a competitively unhealthy degree: the Herfindahl-Hirschman Index is 810 for states and 628 for sectors, both in the 'unconcentrated/competitive' range, even though Maharashtra, Gujarat, and Tamil Nadu together account for roughly 40% of national NVA and the top 5 industries (Basic Metals, Motor Vehicles, Chemicals, Pharmaceuticals, and Machinery) contribute 43.9%. Third, cross-sectional machine-learning models show that at the state level, the number of workers is the strongest predictor of NVA (Gradient Boosting, Leave-One-Out cross-validated $R^2 = 0.930$), whereas at the sector level, fixed capital dominates (Random Forest, LOO-CV $R^2 = 0.815$) — a distinction confirmed independently by SHAP attribution.

A polynomial time-trend model ($R^2 = 0.995$ on the 1989-2024 subset used for modeling) projects national industrial output will reach approximately Rs. 24.0 crore lakh ($\approx$ Rs. 2,401,084,574 lakh) by 2028-29, implying continued double-digit nominal growth, though this projection is a statistical extrapolation of the historical curve — including its sharp post-COVID rebound — rather than a policy-conditioned forecast, and should be read as an indicative upper-range trajectory rather than a point prediction. The report also identifies specific over- and under-performing states and sectors relative to what their capital, labour, and input scale alone would predict, flags an efficiency frontier of capital-to-labour productivity, and segments states and sectors into three natural clusters each via K-Means — providing several concrete, evidence-based starting points for targeted industrial policy.

The analysis is transparent about its limits: the cross-sectional models operate on very small samples (29–36 units), the national forecast uses nominal (non-inflation-adjusted) figures, and several exploratory steps (clustering, the efficiency frontier) are sensitive to feature choice and should be read as indicative rather than confirmatory. These limitations, and what a fuller study would need to resolve them, are detailed in Section 8.

1. Introduction

1.1 Background

The Annual Survey of Industries (ASI) is India's principal source of statistics on the organized (registered) manufacturing sector, conducted by the National Statistical Office under MoSPI. It covers factories registered under the Factories Act, 1948, and is the basis for estimating manufacturing's contribution to India's Gross Value Added. The ASI 2023-24 round — whose reference period runs from 1 April 2023 to 31 March 2024, with fieldwork conducted between November 2024 and June 2025 — provides both a fresh cross-sectional snapshot of the sector and, when read alongside the published annual time series, a 43-year window (1981-82 to 2023-24) into how Indian industry has structurally evolved.

This project uses that window to move beyond a single-year snapshot and ask three connected questions: how has Indian industry grown over four decades; which industries, states, and inputs currently drive that growth;

and how well can standard statistical and machine-learning methods explain and project that growth given the data actually published. The analysis is deliberately built entirely from official ASI tables — no external series (GDP, inflation, exchange rates) are merged in — so every number in this report traces back to a MoSPI-published figure or a model trained directly on one.

1.2 Objectives

The project is organized around five objectives:

1. Trend Assessment — quantify long-run growth in industrial output, net value added, fixed capital (investment), and employment from 1981-82 to 2023-24, and identify structural turning points (e.g. the COVID-19 disruption and recovery).
2. Structural Composition — identify which industries and which states/UTs dominate production and value creation in the 2023-24 reference year, and quantify how concentrated that dominance is.
3. Productivity & Efficiency — quantify capital intensity and labour productivity across industries and states, identify which units sit on the efficiency frontier, and flag where similar inputs are producing markedly different value added.
4. Predictive & Explanatory Modeling — fit a time-series model to project near-term national industrial output, and fit cross-sectional machine-learning models (Ridge, Random Forest, Gradient Boosting) to explain what actually drives NVA across states and across sectors, using model-agnostic explainability (SHAP) to verify the result.
5. Segmentation & Policy Inputs — cluster states and sectors into natural groups using K-Means, and flag specific over- and under-performing states/sectors relative to model expectations, as concrete inputs for regionally or sectorally targeted industrial policy.

1.3 Scope and Coverage

The analysis draws on four ASI 2023-24 published tables:

- Table 1 — Annual Series for Principal Characteristics (1981-82 to 2023-24): 49 characteristics × 43 years, the national time series used for long-run trend and forecasting.
- Table 2 — Principal Characteristics by Major Industry Group (2023-24): 29 industry groups × 16 characteristics, the single-year industry cross-section.
- Table 3 — Principal Characteristics by Major States (2023-24): 36 states/UTs × 15 characteristics, the single-year state cross-section.
- Table 5 — Estimates of Important Characteristics by 3-digit NIC-2008 (2023-24): 77 granular industry codes × 32 characteristics — loaded and validated, though not directly modeled given its lower level of aggregation; noted as an opportunity for future, finer-grained work (see Limitations).

All monetary figures throughout this report are in current prices (Rs. Lakh, nominal) exactly as published by ASI; no inflation adjustment or deflation is applied, consistent with how MoSPI itself publishes these tables.

2. Data and Methodology

2.1 Data Preparation and Validation

Each of the four source tables required custom parsing because ASI publishes them as formatted Excel workbooks rather than tidy data files: Table 1 stores repeating year-blocks that had to be located, sliced, and

merged on a common characteristic label; Table 2 and Table 3 carry a two-row wrapped column header and an 'ALL INDIA' / 'All India' total row that must be split out before per-unit comparison; Table 3 additionally footnotes 'Lakshadweep*' with an asterisk embedded in the state name itself; and Table 5 is stored transposed (characteristics as rows, industry codes as columns) and was transposed back so that, like Table 2, each row is one industry. Footnoted values in Table 1 (marked with a trailing '*' for entries tied to a definitional note) were stripped of the asterisk and coerced to numeric so they would not silently break downstream aggregation.

Every cross-sectional table was validated by summing its constituent units and comparing the sum to the corresponding published All-India / ALL total: Table 2's 29 industries summed to Rs. 1,532,716,610 lakh in Total Output against a published total of Rs. 1,532,716,609 lakh; Table 3's 36 states/UTs summed to the identical Rs. 1,532,716,610 lakh against the same Rs. 1,532,716,609 lakh control; and Table 5's 77 NIC codes plus its 'Others' bucket summed to Rs. 1,532,716,611 lakh in Value of Output against the same control total. In every case the discrepancy is 1–2 lakh — rounding noise, not a data-quality problem — confirming the parsed tables are fit for analysis. Missing values were minimal: 733 cells in Table 1 (expected, since several characteristics were only introduced or redefined partway through the 43-year series), 77 cells in Table 5, and none at all in Table 2 or Table 3.

2.2 Trend Analysis (Table 1, 1981-82 to 2023-24)

Compound annual growth rates (CAGR) were computed for Value of Output, Net Value Added, Fixed Capital, and Number of Workers over the full 43-year span using the standard geometric-mean formula (end value ÷ start value, raised to 1/n years, minus 1). All four series, plus a combined 4-panel view, were plotted on their native (non-log) scale to make the acceleration in later years visually apparent.

2.3 Cross-Sectional Descriptive Analysis (Tables 2 & 3, 2023-24)

For the single reference year 2023-24, industries and states were ranked by Net Value Added, and each unit's share of national NVA was computed. Labour productivity (NVA per worker) and capital intensity (Fixed Capital per worker) were derived for every industry and every state, from which:

- An efficiency frontier was constructed by identifying, for each cross-section, the subset of units for which no other unit achieves both lower-or-equal capital intensity and higher NVA per worker — i.e. the non-dominated set in the (capital intensity, labour productivity) plane.
- Market concentration was measured with the Herfindahl-Hirschman Index (HHI), computed as the sum of squared percentage shares of national NVA across all units in a cross-section (states, and separately sectors), on the conventional 0–10,000 scale where below 1,500 is considered unconcentrated/competitive.

2.4 Predictive Modeling — National Output Forecast

Two related but distinct time-trend models were fitted to national Value of Output, both using Year Index as the sole time variable but differing in scope and covariates:

- An exploratory model (Part 4 of the descriptive analysis) fits simple Linear and Polynomial (degree 2) Regression on Year Index alone, using the full 43-year series (1981-82 to 2023-24), and forecasts 2024-25 through 2028-29.
- A dedicated modeling-stage model ('Model 1') restricts to the 35 years with complete covariate data (1989-90 to 2023-24) and fits Linear and Polynomial (degree 2) Regression using Year Index, Fixed Capital, and Workers as standardized predictors, again forecasting the same five years forward. This is the more methodologically complete of the two and is treated as primary; the exploratory model is retained because it is the one visualized with an explicit confidence band.

In both cases, R^2 on the fitted (training) years is reported, and the higher- R^2 polynomial specification is used for the forward forecast, since the trend visibly curves rather than following a straight line — consistent with an economy whose industrial base is both growing and compounding.

2.5 Cross-Sectional Machine-Learning Models (Tables 2 & 3)

To identify what drives NVA across states and, separately, across sectors, three regression models were trained on each cross-section using Fixed Capital, Workers, Total Input, and Number of Factories as predictors of Net Value Added:

- Ridge Regression ($\alpha = 10$) — a regularized linear baseline, robust to the strong multicollinearity among the four predictors (see Section 2.6).
- Random Forest Regressor (300 trees, max depth 5) — a non-linear ensemble capable of capturing interaction effects.
- Gradient Boosting Regressor (300 estimators, max depth 3, learning rate 0.05) — a sequential boosting ensemble, typically the highest-capacity of the three.

All predictors were standardized (zero mean, unit variance) before fitting. Because each cross-section has only 29–36 observations, Leave-One-Out Cross-Validation (LOO-CV) — refitting the model with each single unit held out in turn and predicting it — was used as the primary evaluation metric rather than a k-fold split, since LOO-CV is the appropriate validation strategy when the sample is too small to spare a separate test fold. For each cross-section, the model with the highest LOO-CV R^2 was used to generate final predictions, and each unit's residual (Actual NVA – Predicted NVA) was used to flag it as an over-performer (producing more value added than its inputs alone would predict) or under-performer (producing less).

2.6 Correlation, Feature Importance, and SHAP Explainability

A Pearson correlation matrix was computed among Fixed Capital, Workers, Total Input, Number of Factories, and Net Value Added, separately for states and for sectors, both to characterize how the four industrial inputs co-move and to justify the choice of tree-based models over plain linear regression (high multicollinearity among predictors inflates and destabilizes linear coefficients but does not impair tree-based splits). Built-in feature importances (mean decrease in impurity) were extracted from the fitted Random Forest and Gradient Boosting models for both cross-sections. As an independent check, SHAP (SHapley Additive exPlanations) values were computed for the Gradient Boosting models — a game-theoretic attribution method that assigns each prediction's deviation from the average to individual features — and both the mean absolute SHAP value per feature and the full per-observation SHAP distribution (beeswarm) were examined.

2.7 Segmentation — K-Means Clustering

States and sectors were separately clustered using K-Means on their standardized Fixed Capital, Workers, Total Input, and Number of Factories values. The number of clusters (k) was chosen via the elbow method (inertia versus k , for $k = 1$ to 6) for each cross-section, and the resulting clusters were visualized in two-dimensional space via Principal Component Analysis (PCA) projection of the same four standardized features.

2.8 Software and Reproducibility

All modeling was implemented in Python using pandas for data preparation, scikit-learn for Ridge/Random Forest/Gradient Boosting regression and StandardScaler, SciPy/statsmodels-equivalent routines for correlation, a SHAP-compatible explainer for the Gradient Boosting attributions, and Matplotlib for all visualizations. A fixed random_state (42) was used for every stochastic model to ensure the reported figures are exactly reproducible from the underlying notebook.

3. Results and Analysis

3.1 National Industrial Trends, 1981-82 to 2023-24

Over the full 43-year series, India's Value of Output grew at a compound annual rate of 10.51%, Net Value Added at 9.72%, and Fixed Capital (investment) at 12.35% — all nominal — while the number of workers grew at only 2.25% per year. Output and capital thus grew roughly 4.7 times faster than employment, indicating that four decades of industrial growth have been driven predominantly by capital deepening rather than by expanding the workforce.

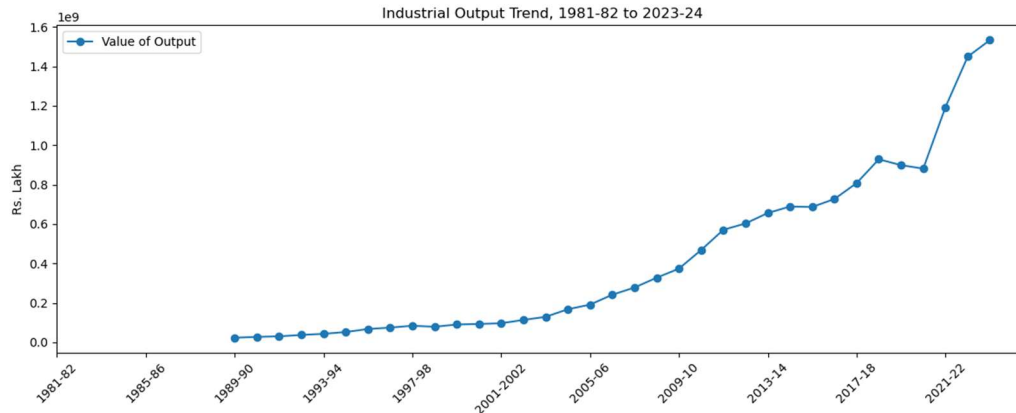


Figure 1: Industrial Output (Value of Output) trend, 1981-82 to 2023-24.

The series is close to flat through the 1980s and 1990s, begins a visible acceleration from the mid-2000s, and shows a sharp dip in 2020-21 followed by the steepest single-year recovery of the entire 43-year series by 2021-22 to 2023-24. This pattern — slow early growth, mid-2000s liberalization-era acceleration, a COVID-19 shock, and a V-shaped rebound — is the clearest visual evidence for Objective 1: industrial output growth has not been steady, it has compounded, which is exactly why a polynomial (not linear) trend model fits it best in Section 3.4.

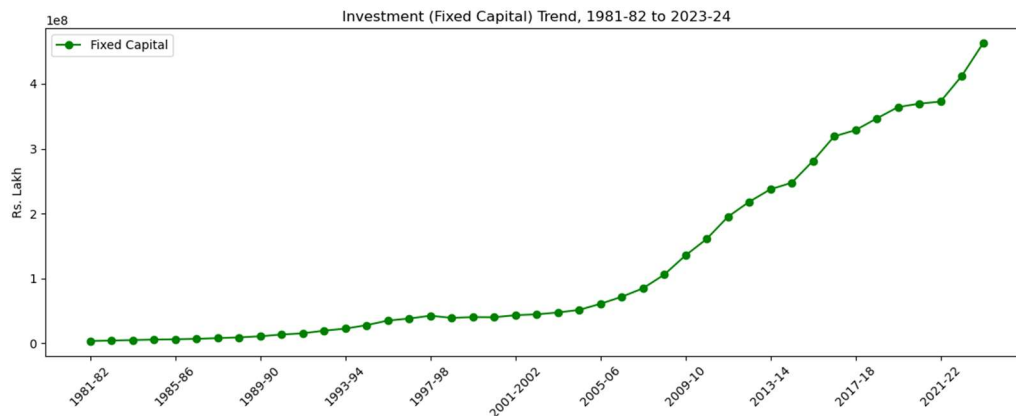


Figure 2: Investment (Fixed Capital) trend, 1981-82 to 2023-24.

Fixed Capital shows the same late-acceleration shape as output but rises even more steeply after 2010-11, consistent with its higher 12.35% CAGR versus output's 10.51%. Because Fixed Capital is outpacing Value of Output over time, the capital-to-output ratio in Indian manufacturing has been rising — each unit of output now requires proportionally more invested capital than it did in the 1990s, a structural shift toward more capital-intensive production techniques with direct implications for how policy should think about investment incentives versus employment-generation schemes.

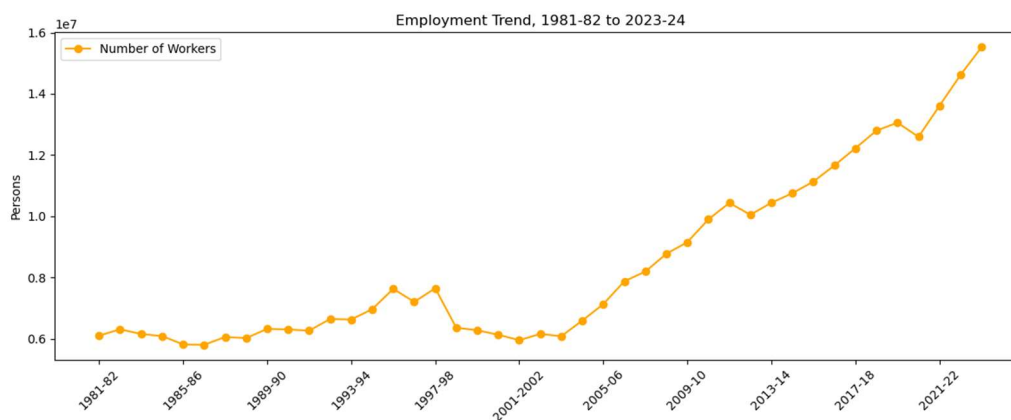


Figure 3: Employment (Number of Workers) trend, 1981-82 to 2023-24.

Employment is visibly the most volatile of the four series relative to its own scale — it dips noticeably in the mid-to-late 1990s and again sharply in 2020-21, but its long-run slope is far shallower than output or capital's. The workforce roughly doubled from 6.1 million to 15.5 million workers over 43 years while output grew nearly 60-fold, which is the clearest numeric confirmation of Objective 1's central finding: Indian industrial growth has been capital-led, and job creation has lagged well behind value creation, a gap directly relevant to any policy discussion of jobless growth in manufacturing.

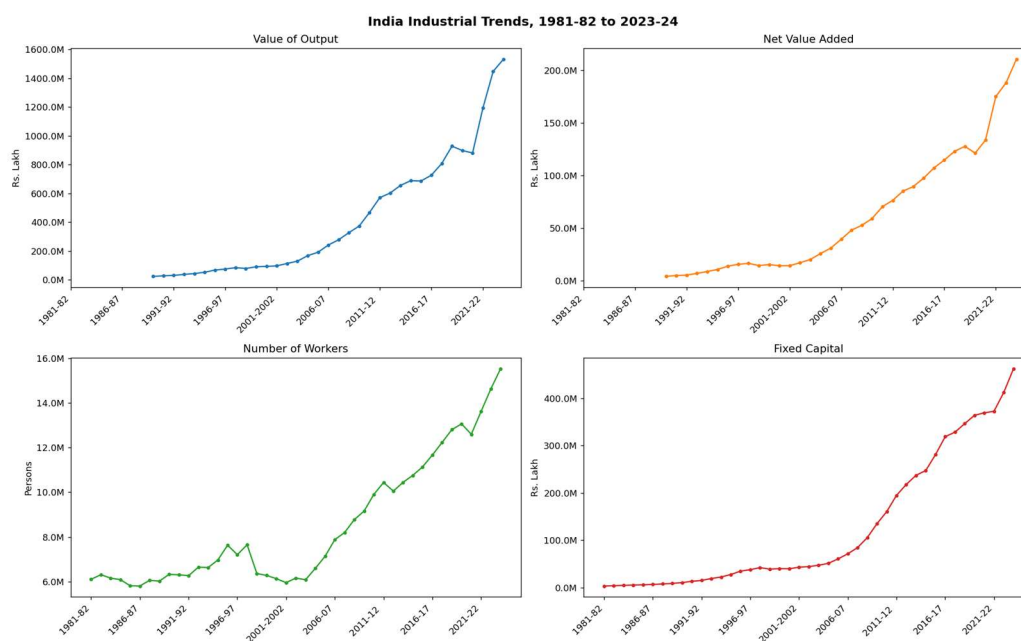


Figure 4: Combined view of Value of Output, Net Value Added, Number of Workers, and Fixed Capital, 1981-82 to 2023-24.

Viewing all four series on a shared timeline makes the divergence explicit: Output, NVA, and Fixed Capital all show the same steep late-stage acceleration and near-identical COVID-dip-and-recovery shape, while Workers traces a visibly flatter, choppier path throughout. The near-perfect co-movement of Output, NVA, and Fixed Capital (all rising and dipping together) versus the decoupled Workers series is the single most policy-relevant chart in the trends section: it shows in one view that the sources of industrial growth (capital, and the output/value it generates) have been moving together, while the beneficiary population of that growth in direct employment terms has not kept pace.

3.2 Industry-Wise Performance, 2023-24

In the 2023-24 reference year, the top 5 industries by Net Value Added — Basic Metals (Rs. 23,415,506 lakh), Motor Vehicles/Trailers/Semi-Trailers (Rs. 18,970,269 lakh), Chemicals and Chemical Products (Rs. 18,450,212 lakh), Pharmaceuticals/Medicinal Chemical/Botanical Products (Rs. 16,077,146 lakh), and Machinery and Equipment n.e.c. (Rs. 15,468,269 lakh) — together account for 43.9% of total industrial NVA. At the other end, Salt Production by Evaporation of Sea Water (Rs. 63,692 lakh), Publishing Activities (Rs. 262,369 lakh), Waste Collection/Treatment & Disposal (Rs. 267,628 lakh), Repair and Installation of Machinery and Equipment (Rs. 445,162 lakh), and Wood and Products of Wood and Cork except Furniture (Rs. 661,945 lakh) are the five smallest contributors.

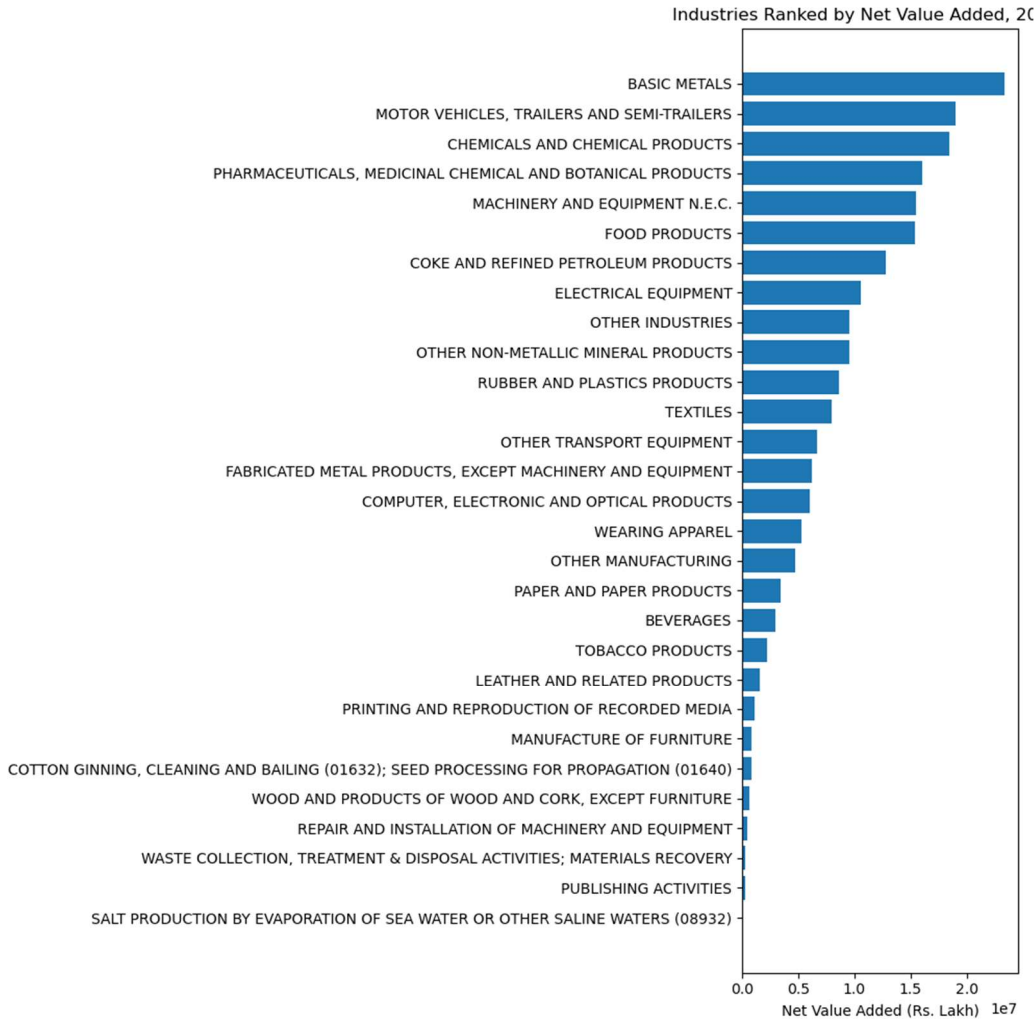


Figure 5: All 29 industries ranked by Net Value Added, 2023-24 (initial exploratory ranking).

This first-pass ranking, produced during the exploratory phase of the analysis, already reveals the steep drop-off from Basic Metals at the top to the long tail of smaller industries — the bar lengths shrink roughly geometrically rather than linearly moving down the list, an early visual signal of the concentration that Section 3.7's HHI calculation later quantifies precisely (top-5 share of 43.9%).

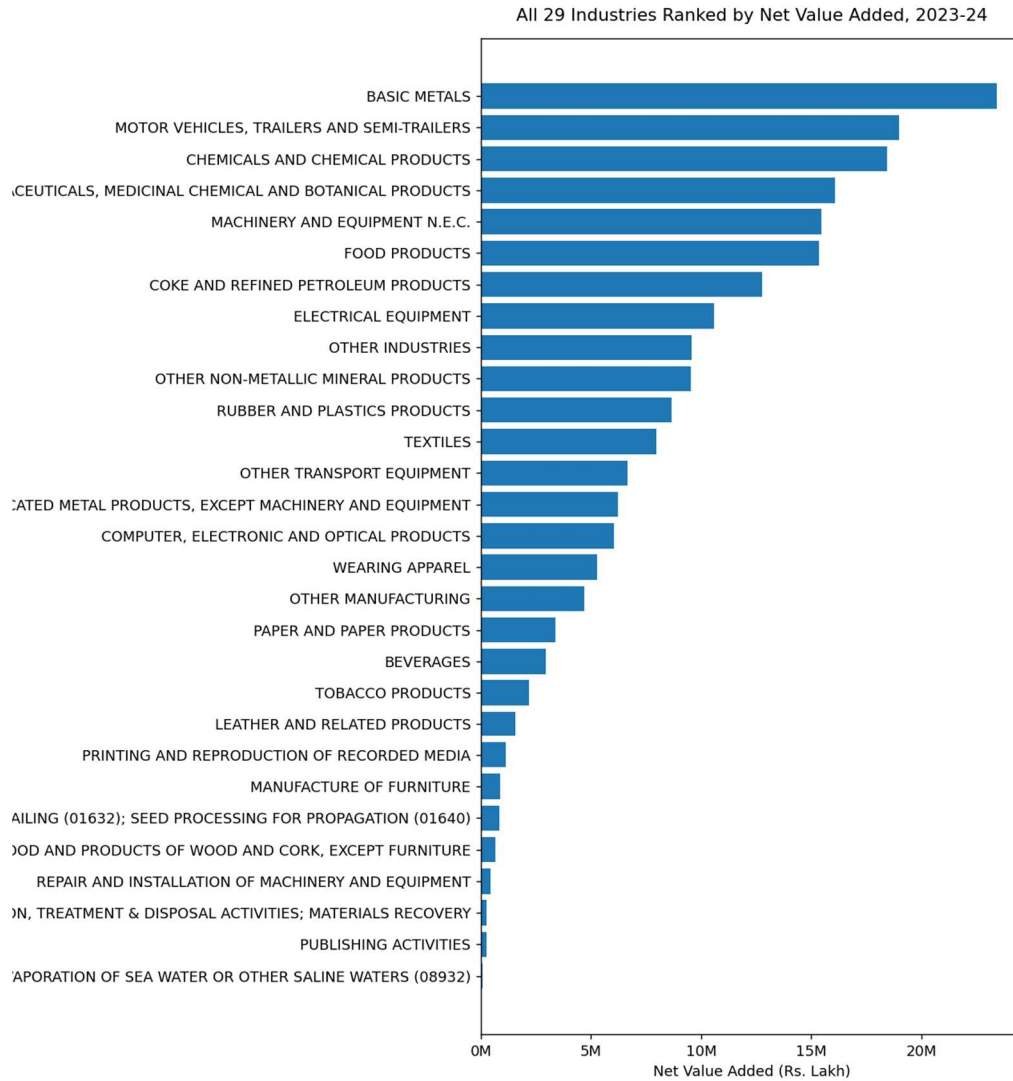
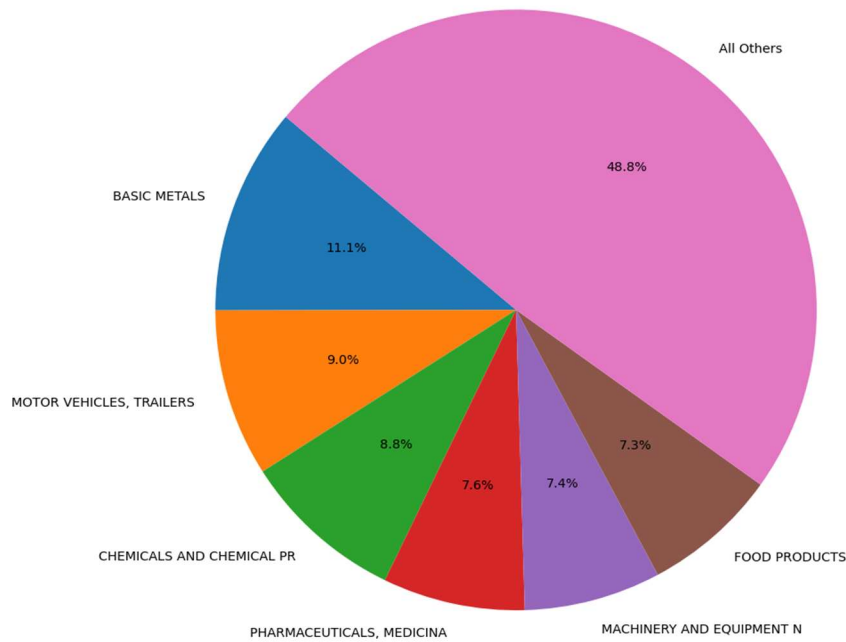


Figure 6: All 29 industries ranked by Net Value Added, 2023-24 (refined final view, full industry names).

This refined version confirms the same ranking with full industry labels, making the composition of India's industrial value-add legible at a glance: heavy/basic manufacturing (Basic Metals), transport manufacturing (Motor Vehicles, Other Transport Equipment), and chemicals-adjacent industries (Chemicals, Pharmaceuticals) dominate the top of the list, while smaller, more traditional or informal-adjacent activities (wood products, salt production, publishing) sit at the bottom. For policy planning this ranking is the starting point for any sector-specific intervention — it identifies exactly which industries carry the most weight in national industrial value added and would therefore have the largest absolute national impact per unit of policy effort directed at them.

Share of National Net Value Added by Industry, 2023-24

**Figure 7:** *Share of national Net Value Added by industry, 2023-24 — top 6 industries versus 'All Others'.*

Even the leading industry, Basic Metals, accounts for only 11.1% of national industrial NVA, and the combined top 6 (Basic Metals, Motor Vehicles, Chemicals, Pharmaceuticals, Machinery, and Food Products) still leave 48.8% — just under half — in the 'All Others' category spread across the remaining 23 industries. This is a useful counterweight to Figure 5/6: while the top industries are large in absolute terms, no single industry has anything close to a dominant share, which is exactly why the HHI concentration measure in Section 3.7 classifies sector-level NVA as competitive/unconcentrated (HHI = 628) rather than oligopolistic.

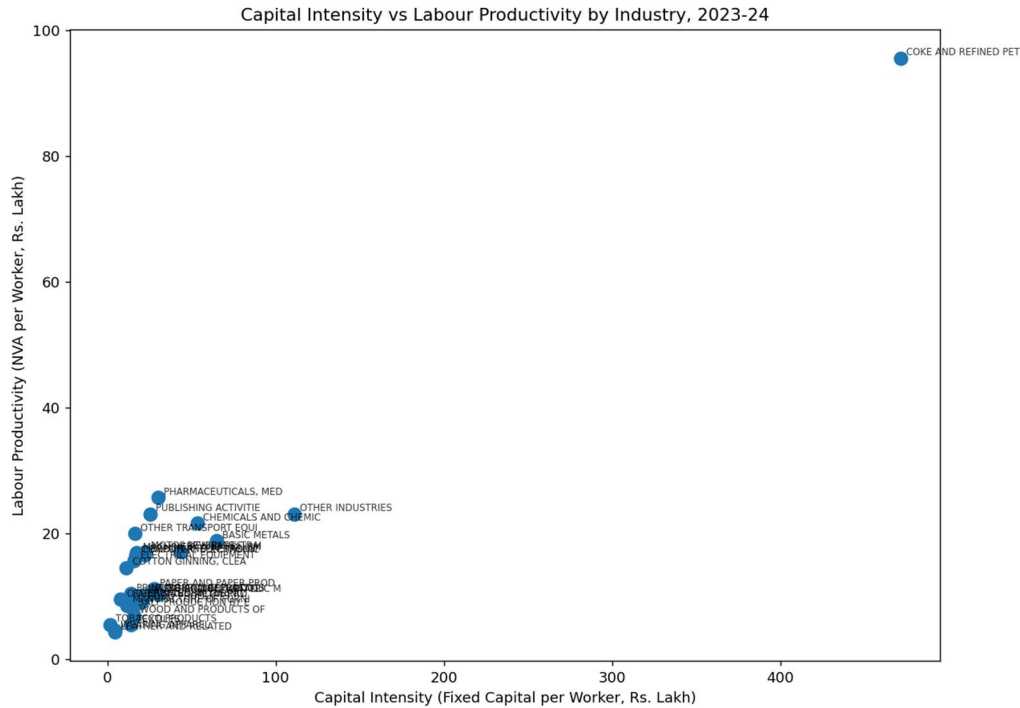


Figure 8: Capital intensity (Fixed Capital per worker) versus labour productivity (NVA per worker) across all 29 industries, 2023-24.

Coke and Refined Petroleum Products is a clear outlier on both axes — Rs. 471.5 lakh of fixed capital per worker and Rs. 95.6 lakh of NVA per worker, roughly 4–5 times higher than any other industry on either measure, consistent with refining being an extremely capital-intensive, highly automated process with very high value per employee. The remaining 28 industries cluster in a much lower range (mostly under Rs. 70 lakh capital per worker and under Rs. 27 lakh NVA per worker), with Pharmaceuticals, Publishing, and Other Industries standing out as achieving comparatively high labour productivity without correspondingly high capital intensity — precisely the profile the efficiency-frontier analysis in Section 3.7 formalizes.

3.3 State-Wise Performance, 2023-24

Maharashtra (Rs. 34,145,981 lakh), Gujarat (Rs. 29,099,304 lakh), and Tamil Nadu (Rs. 21,395,650 lakh) are the three largest contributors to national industrial NVA, together accounting for roughly 40% of the national total — a concentration consistent with these three states' long-established industrial and port infrastructure. At the other end, several smaller states and Union Territories (Mizoram, A & N Islands, Ladakh, Lakshadweep) report NVA below Rs. 2,000 lakh, reflecting their much smaller registered-manufacturing base.

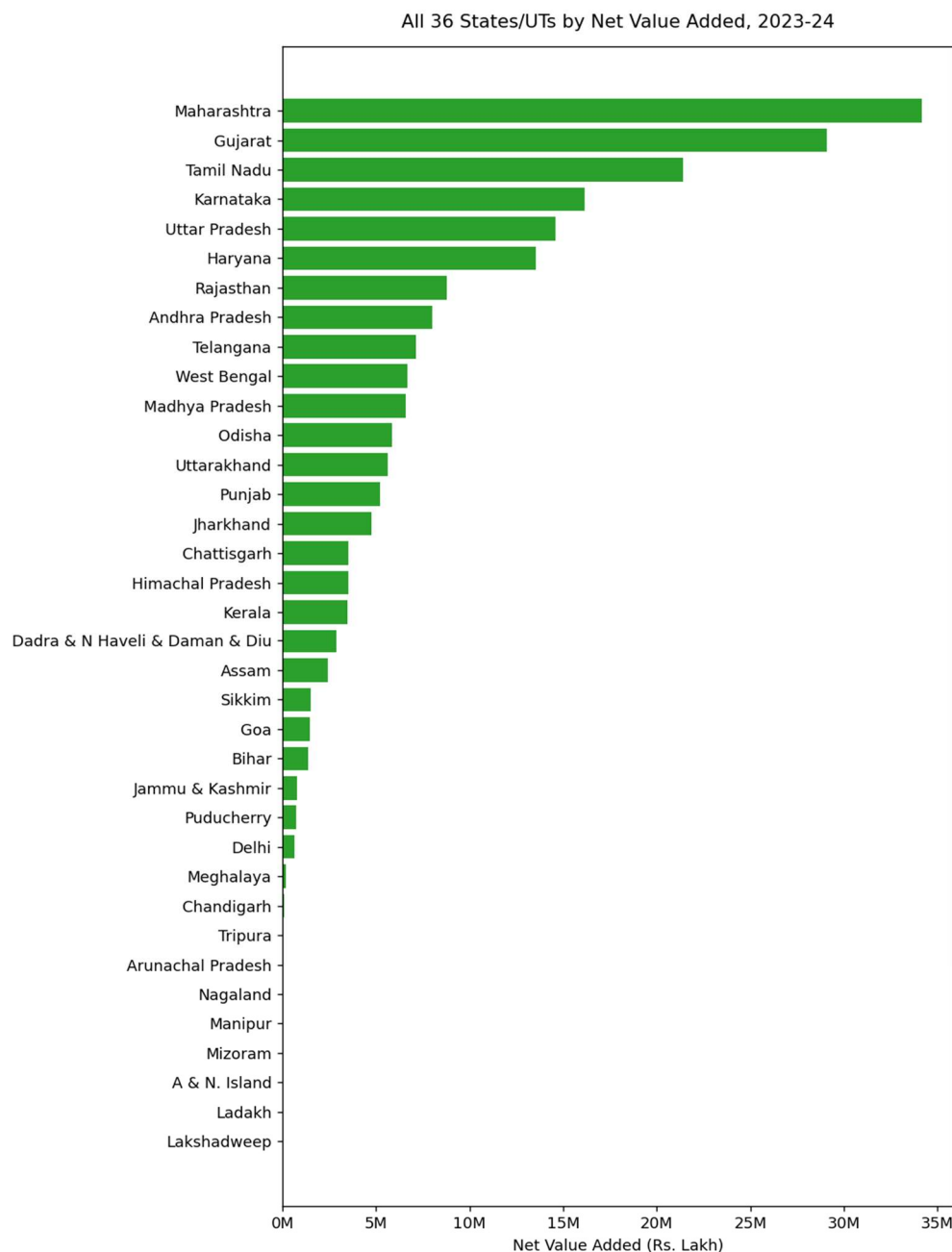


Figure 9: All 36 states and Union Territories ranked by Net Value Added, 2023-24.

The ranking shows a steep concentration at the top (Maharashtra, Gujarat, Tamil Nadu, Karnataka, Uttar Pradesh, and Haryana together clearly dominating the chart) followed by a long tail of mid-sized industrial states (Rajasthan through Bihar) and a cluster of very small contributors at the bottom, several of which are Union Territories or northeastern/hill states with limited industrial land and factory density. This state-level concentration is the geographic mirror of the industry-level concentration in Section 3.2, and together they define the two axes — which industries, and which states — that any regionally or sectorally targeted industrial policy needs to jointly consider; the cross-sectional models in Section 3.5 build directly on this state ranking to ask which states are converting their capital and labour into NVA more or less efficiently than their peers.

3.4 National Output Forecast

Two related polynomial time-trend models were fitted. The exploratory model, fit on the full 43-year series with Year Index as the only predictor, achieves $R^2 = 0.9738$ and is the one visualized below with an explicit forecast band. The primary modeling-stage model ('Model 1'), fit on the 35-year 1989-2024 subset using Year Index, Fixed Capital, and Workers as standardized predictors, achieves a considerably tighter fit — $R^2 = 0.9954$ for the polynomial specification versus $R^2 = 0.9658$ for linear regression alone — confirming that adding capital and labour as explicit covariates, not just letting time proxy for everything, meaningfully improves the national output model.

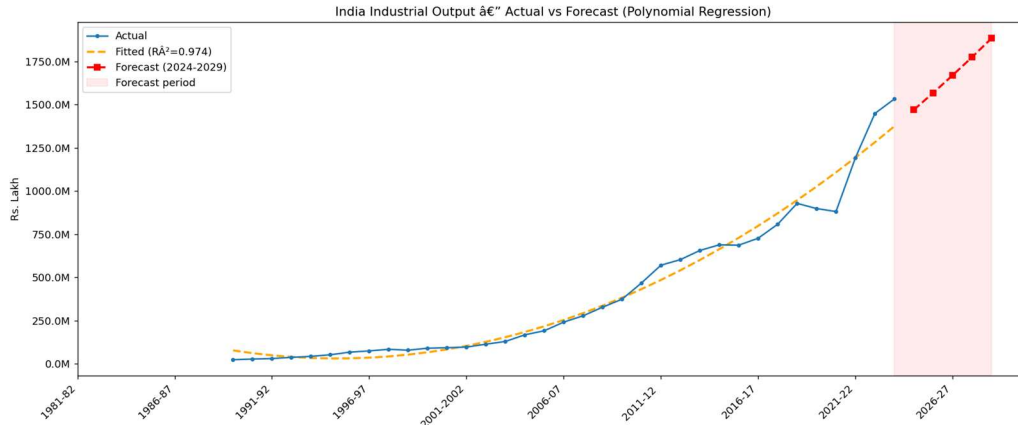


Figure 10: India's industrial output (Value of Output) — actual values, fitted polynomial trend ($R^2 = 0.974$), and 2024-29 forecast with shaded forecast period.

The fitted orange curve tracks the actual blue series closely across the full history, including its long, slow early decades and its late-stage acceleration — visual confirmation of the model's high R^2 . The red forecast segment (2024-25 to 2028-29) extrapolates that accelerating curve forward to a projected Rs. 1,883,921,611 lakh by 2028-29 under this exploratory specification. It is worth reading this forecast together with the primary Model 1 figures below: because Model 1 additionally conditions on Fixed Capital and Workers rather than time alone, and because the fitted curve is being extrapolated through the unusually steep post-COVID recovery years, both forecasts should be treated as upper-range, trend-continuation projections rather than a single best estimate — a caveat the notebook's own model notes make explicit.

Table 1: Model 1 — 5-Year National Output Forecast (Rs. Lakh)

Year	Linear Regression	Polynomial Regression (deg. 2)
2024-25	1,436,975,832	1,667,731,437
2025-26	1,531,347,761	1,832,949,940
2026-27	1,630,412,975	2,010,074,543
2027-28	1,734,418,993	2,199,411,020
2028-29	1,843,626,741	2,401,084,574

Model 1 (fit on 1989-90 to 2023-24 with Fixed Capital and Workers as covariates) has $R^2 = 0.9658$ for the linear specification and $R^2 = 0.9954$ for the polynomial specification; the higher- R^2 polynomial forecast is used above and projects output reaching approximately Rs. 2,401,084,574 lakh by 2028-29 — a considerably steeper trajectory than the time-only exploratory model, reflecting the additional lift from projecting continued capital and labour growth alongside time.

3.5 Cross-Sectional Machine-Learning Analysis: What Drives NVA?

Two independent cross-sectional models were built — one across all 36 states/UTs, one across all 29 industries — each predicting Net Value Added from Fixed Capital, Workers, Total Input, and Number of Factories. Model

performance is summarized below; in both cases, the highest-capacity tree-based model achieves near-perfect in-sample fit, but the Leave-One-Out cross-validated R^2 (the honest, out-of-sample-equivalent metric for these small samples) is meaningfully lower — a healthy and expected gap that reflects genuine model uncertainty rather than a modeling error.

Table 2: State Cross-Sectional Model Performance ($n = 36$)

Model	R^2 (in-sample)	R^2 (LOO-CV)	MAE (Rs. Lakh)
Ridge Regression	0.9344	0.9007	1,207,788
Random Forest	0.9900	0.9186	424,262
Gradient Boosting	1.0000	0.9301	4,761

Gradient Boosting is the best-performing state-level model by LOO-CV R^2 (0.9301) and is used for the residual analysis below.

Table 3: Sector Cross-Sectional Model Performance ($n = 29$)

Model	R^2 (in-sample)	R^2 (LOO-CV)	MAE (Rs. Lakh)
Ridge Regression	0.7989	0.7408	2,232,024
Random Forest	0.9763	0.8151	683,684
Gradient Boosting	1.0000	0.7968	1,756

Random Forest is the best-performing sector-level model by LOO-CV R^2 (0.8151) — notably, Gradient Boosting achieves the highest in-sample fit in both cross-sections (as expected of the highest-capacity model) but is not the best out-of-sample performer for sectors, a useful reminder that in-sample R^2 alone would have led to the wrong model choice here.

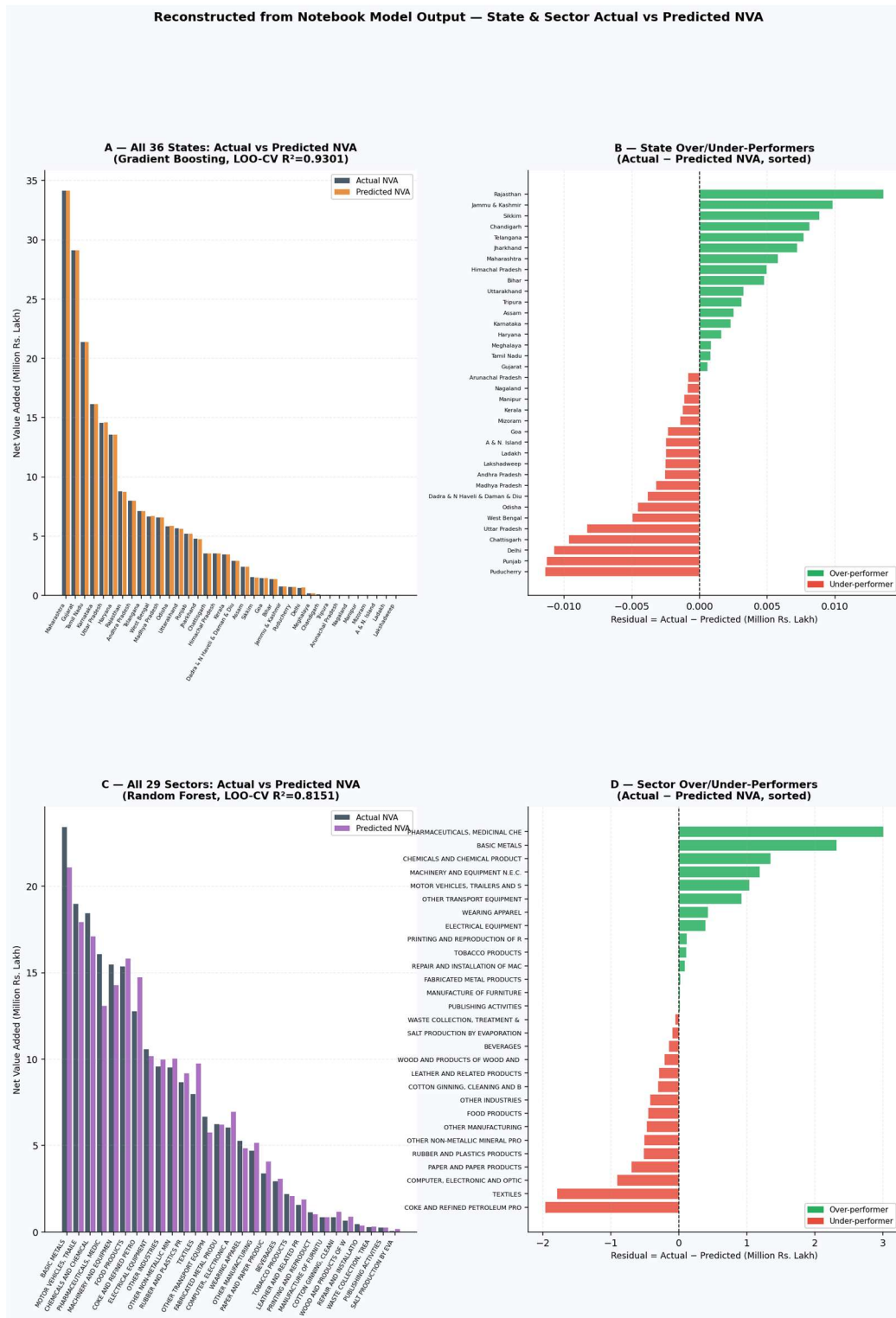


Figure 11: State and sector actual vs. predicted NVA (best model per cross-section), and sorted over/under-performer residuals for both.

Panels A and C show that the best model for each cross-section (Gradient Boosting for states, Random Forest for sectors) tracks the overall shape of the actual NVA distribution closely across every state and sector, including the very large gap between the top few units and the long tail — exactly what the high LOO-CV R^2 values in Tables 2 and 3 would predict. Panels B and D isolate the residuals: among states, Rajasthan, Jammu & Kashmir, and Sikkim are the clearest over-performers (delivering more NVA than their capital/labour/input scale alone

predicts), while Puducherry, Punjab, and Delhi are the clearest under-performers; among sectors, Pharmaceuticals, Basic Metals, and Chemicals over-perform their input-based prediction, while Coke and Refined Petroleum Products, Textiles, and Computer/Electronic/Optical Products under-perform. For policy purposes, over-performers are worth studying as models of efficient input use (their input mix or industry composition converts to value added particularly well), while under-performers represent states or sectors where capital and labour are present but not converting into value added as effectively as elsewhere — i.e. where the same inputs, if better utilized, could support meaningfully more output without new capital investment.

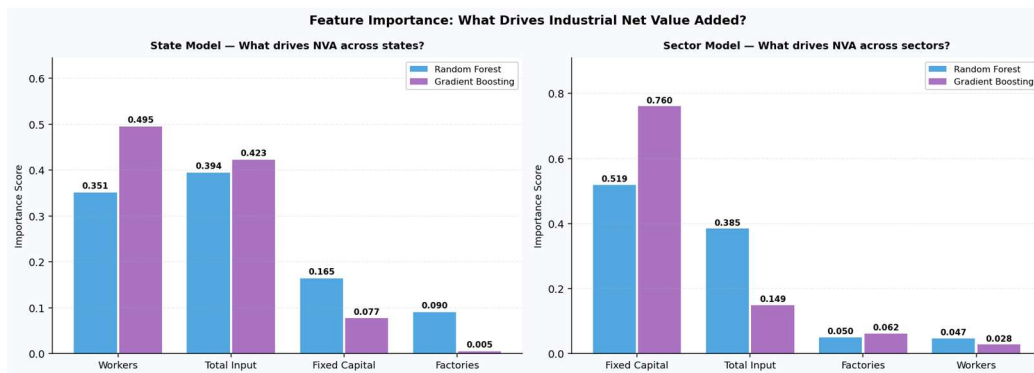


Figure 12: Feature importance for state-level and sector-level NVA models (Random Forest vs. Gradient Boosting).

The two panels tell a genuinely different story for states versus sectors. For states (left), Workers is the single most important feature under Gradient Boosting (importance 0.495, versus 0.351 under Random Forest), narrowly ahead of Total Input (0.423 / 0.394), while Fixed Capital and Number of Factories matter comparatively little. For sectors (right), the ranking flips: Fixed Capital dominates (0.760 under Gradient Boosting, 0.519 under Random Forest), well ahead of Total Input, with Factories and Workers contributing little. This is one of the report's central findings: value added across Indian states scales primarily with headcount, while value added across industry sectors scales primarily with invested capital — a distinction with direct implications for policy design, since a state-focused employment-generation lever and a sector-focused capital-incentive lever are pulling on two different, empirically validated mechanisms.

3.6 Correlation Structure

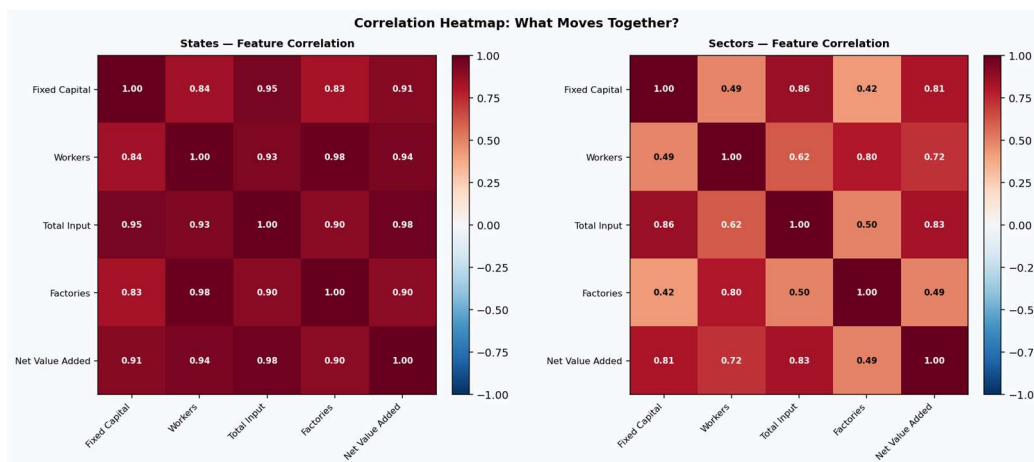


Figure 13: Pearson correlation matrix among Fixed Capital, Workers, Total Input, Number of Factories, and Net Value Added — states (left) and sectors (right).

Among states, every pair of the four input variables is very strongly correlated (all above 0.83, with Workers and Factories reaching 0.98), and each is correlated with NVA at 0.90 or above — states that are large on one input

dimension tend to be large on all of them and on NVA, reflecting that a handful of states (Maharashtra, Gujarat, Tamil Nadu, and a few others) are simply larger industrial economies across every measure simultaneously. Among sectors, the correlations are present but visibly weaker and more varied (Fixed Capital-to-Workers is only 0.49, and Factories-to-NVA only 0.49), showing that sectors are more differentiated — some are large in capital but not workers (like Coke and Refined Petroleum), others large in workers but not capital (more labour-intensive manufacturing). This is precisely why the state and sector feature-importance rankings in Figure 12 diverge: with states, the input variables are so collinear that whichever one the model leans on captures nearly the same information; with sectors, the inputs are more independent, so the model's clear preference for Fixed Capital is a more informative, less coincidental signal.

3.7 Productivity, Efficiency, and Market Concentration

Table 4: Top 10 States by Labour Productivity (NVA per Worker)

State	NVA/Worker (Rs. Lakh)	Capital/Worker (Rs. Lakh)	Input Efficiency
Sikkim	73.42	25.22	0.963
Jharkhand	25.00	55.82	0.231
Goa	22.33	26.91	0.281
Himachal Pradesh	20.50	28.62	0.285
Odisha	18.61	117.19	0.132
Maharashtra	18.46	29.86	0.187
Madhya Pradesh	17.13	63.19	0.179
Puducherry	16.96	16.55	0.231
Karnataka	16.68	29.14	0.202
Uttarakhand	16.05	18.03	0.257

Table 5: Top 10 Sectors by Labour Productivity (NVA per Worker)

Industry	NVA/Worker (Rs. Lakh)	Capital/Worker (Rs. Lakh)	Input Efficiency
Coke and Refined Petroleum Products	95.60	471.49	0.075
Pharmaceuticals, Medicinal Chemical & Botanical Products	25.74	30.26	0.472
Publishing Activities	23.14	25.34	0.475
Other Industries	23.11	111.10	0.286
Chemicals and Chemical Products	21.67	53.59	0.180
Other Transport Equipment	20.03	16.47	0.259
Basic Metals	18.84	64.82	0.121
Beverages	17.11	43.59	0.241
Motor Vehicles, Trailers and Semi-Trailers	17.10	22.65	0.177
Machinery and Equipment n.e.c.	16.93	17.47	0.259

At the bottom of the state ranking, Mizoram (1.71), Nagaland (1.85), Tripura (1.87), Manipur (2.07), and Ladakh (2.08) report the lowest NVA per worker — high relative headcount against comparatively low industrial value creation, consistent with these states' smaller, less capital-intensive registered manufacturing base.

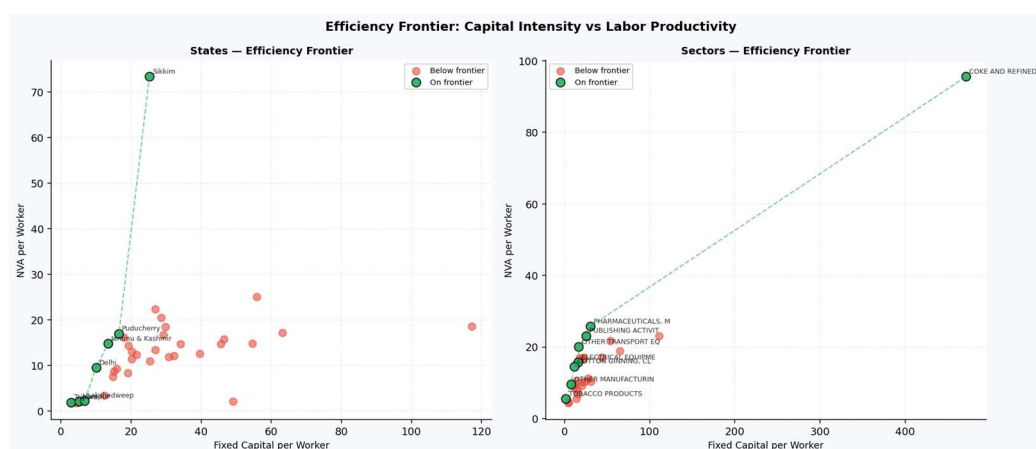


Figure 14: Efficiency frontier: capital intensity versus labour productivity for states (left) and sectors (right), with non-dominated ('on frontier') units highlighted.

Sikkim sits dramatically above the rest of the state frontier — a small state achieving Rs. 73.4 lakh NVA per worker at only Rs. 25.2 lakh capital per worker, the single most capital-efficient state in the country by this measure. Delhi, Jammu & Kashmir, and Puducherry form the lower-capital segment of the state frontier, while most other states cluster well inside the frontier, meaning they are producing less NVA per worker than some other state already demonstrates is achievable at equal or lower capital intensity. On the sector side, Coke and Refined Petroleum Products anchors the far end of the frontier at extreme capital intensity, while Pharmaceuticals, Publishing, Other Transport Equipment, Electrical Equipment, and Tobacco Products define the frontier at much lower capital levels. The direct policy reading: any state or sector lying inside (below) the frontier has empirical proof, from a peer unit, that materially higher labour productivity is achievable without first raising capital intensity — pointing toward operational, organizational, or skill-based interventions rather than capital subsidy alone.

Table 6: Concentration of Industrial NVA — Top 10 States and Sectors

State-level NVA concentration: HHI = 810 (unconcentrated / competitive); top-5 state share of national NVA = 54.9%. Sector-level NVA concentration: HHI = 628 (unconcentrated / competitive); top-5 sector share of national NVA = 43.9%.

Top 10 States	Share of National NVA (%)
Maharashtra	16.2
Gujarat	13.8
Tamil Nadu	10.2
Karnataka	7.7
Uttar Pradesh	6.9
Haryana	6.4
Rajasthan	4.2
Andhra Pradesh	3.8
Telangana	3.4
West Bengal	3.2
Top 10 Sectors	Share of National NVA (%)
Basic Metals	11.1
Motor Vehicles, Trailers and Semi-Trailers	9.0

Top 10 States	Share of National NVA (%)
Chemicals and Chemical Products	8.8
Pharmaceuticals, Medicinal Chemical, Botanical	7.6
Machinery and Equipment n.e.c.	7.4
Food Products	7.3
Coke and Refined Petroleum Products	6.1
Electrical Equipment	5.0
Other Industries	4.6
Other Non-Metallic Mineral Products	4.5

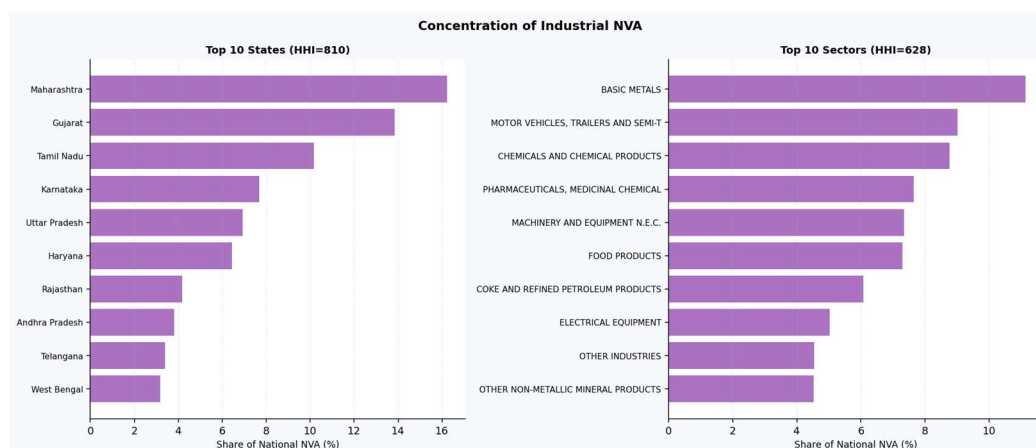


Figure 15: Share of national NVA held by the top 10 states (left) and top 10 sectors (right), 2023-24.

Both panels show a smooth, gradually declining share as one moves down the ranking rather than a sharp cliff after the leader — Maharashtra's 16.2% state-level lead and Basic Metals' 11.1% sector-level lead are both meaningful but far from dominant. This gradual decline is the visual reason both HHI values (810 for states, 628 for sectors) land well inside the conventional 'unconcentrated' threshold of 1,500: national industrial value added, while unevenly distributed, is not captured by any small oligopoly of states or industries. This is a genuinely reassuring structural finding for competition and regional-balance policy, even as the top-5 shares (54.9% for states, 43.9% for sectors) show there is still real room to broaden the industrial base further.

3.8 Structural Segmentation: K-Means Clustering

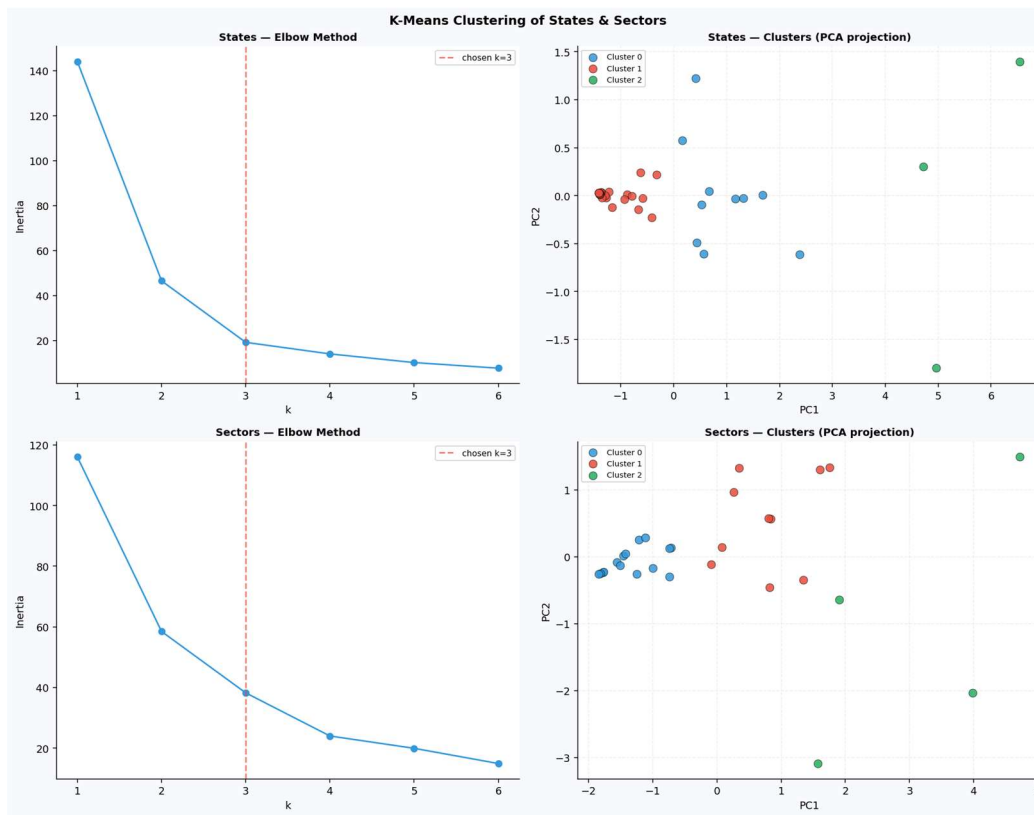


Figure 16: K-Means clustering of states (top) and sectors (bottom): elbow method for choosing k (left) and resulting clusters in PCA-projected space (right).

Both elbow plots show inertia dropping sharply from $k=1$ to $k=3$ and flattening substantially thereafter, supporting $k=3$ as a reasonable, data-driven choice for both cross-sections. Among states, Cluster 2 isolates just 3 states — Maharashtra, Gujarat, and Tamil Nadu — the same three large industrial economies identified repeatedly throughout this report, cleanly separated from a mid-tier Cluster 0 of 10 states (including Karnataka, Uttar Pradesh, Haryana, Rajasthan, Andhra Pradesh, and Telangana) and a large Cluster 1 of 23 smaller states and Union Territories. Among sectors, Cluster 2 similarly isolates 4 heavyweight industries (Basic Metals, Chemicals, Food Products, Coke and Refined Petroleum), Cluster 1 groups 10 mid-scale industries (including Motor Vehicles, Pharmaceuticals, Machinery, and Electrical Equipment), and Cluster 0 holds 15 smaller-scale sectors. In both cases the clustering independently recovers, from raw input scale alone with no reference to NVA, essentially the same 'large / mid / small' segmentation this report has been building toward through rankings, HHI, and the frontier analysis — a useful cross-check that the earlier descriptive findings reflect real structural groupings rather than an artifact of how the data happened to be ranked.

3.9 Model Explainability: SHAP Analysis

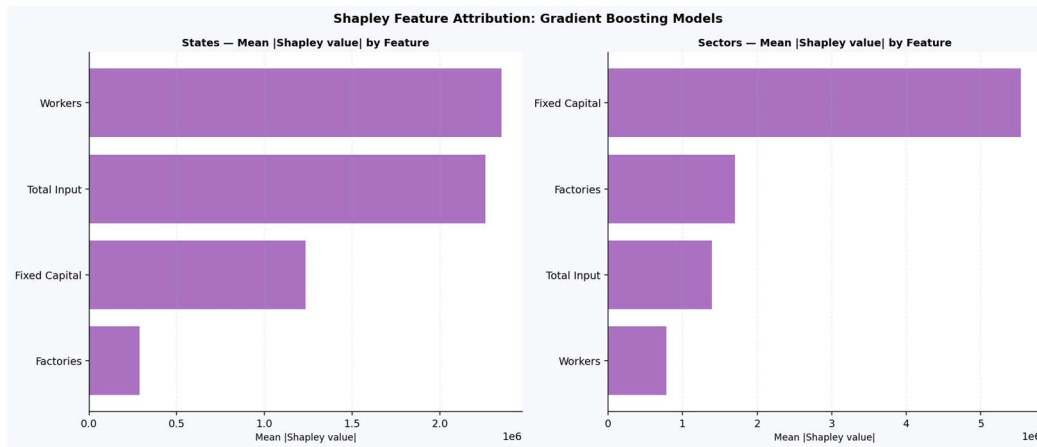


Figure 17: Mean absolute SHAP value by feature for the Gradient Boosting models — states (left) and sectors (right).

SHAP attribution, a method grounded in cooperative game theory rather than simple impurity reduction, independently confirms the built-in feature-importance ranking from Figure 12: Workers carries the largest mean absolute SHAP value for the state model, while Fixed Capital dominates for the sector model, with Total Input a consistent secondary contributor in both. That two different explainability methods — one based on how much each feature reduces prediction error inside the trees, the other based on each feature's marginal contribution across all possible coalitions of features — arrive at the same ranking substantially strengthens confidence that 'Workers drive state-level NVA, Fixed Capital drives sector-level NVA' is a genuine structural finding rather than an artifact of one particular algorithm.

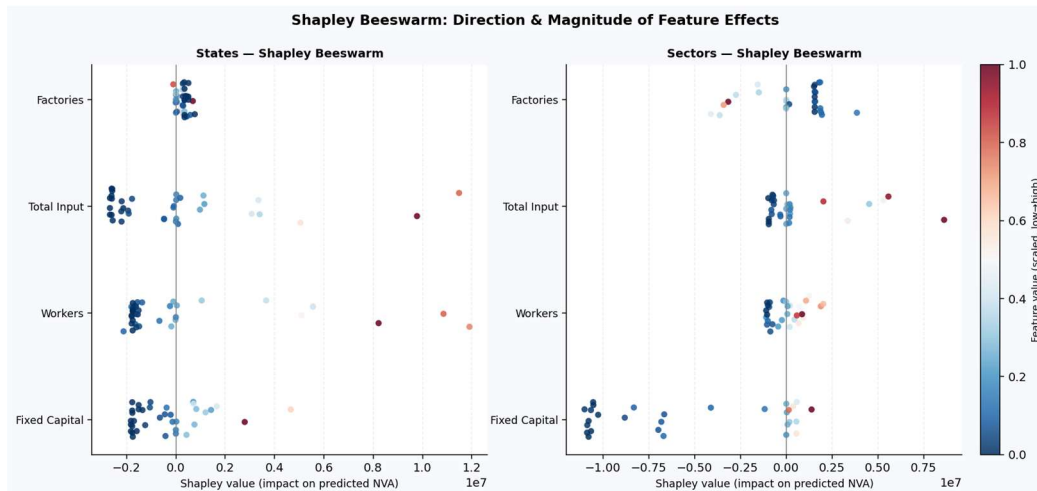


Figure 18: SHAP beeswarm plot: direction and magnitude of each feature's effect on predicted NVA, per individual state/sector observation.

Beyond confirming which feature matters most, the beeswarm shows how: for both states and sectors, high feature values (dark red/orange points, e.g. a state with many workers, or a sector with high fixed capital) are associated with clearly positive SHAP values (pushing predicted NVA up), while low feature values (blue points) cluster near zero or push predictions down — a monotonic, intuitive relationship with no evidence of a feature 'reversing sign' at extreme values. The handful of far-right outlier points in the Total Input and Workers rows (both panels) correspond to the largest states and sectors (Maharashtra/Gujarat/Tamil Nadu on the state side, Basic Metals/Chemicals on the sector side) whose sheer scale contributes an outsized, individually identifiable positive push to their own predicted NVA — consistent with everything this report has shown about industrial concentration among a small set of leading states and sectors.

4. Future Growth Predictions

This section draws together the modeling results from Section 3.4 into a dedicated forward-looking assessment, since projecting near-term industrial growth was one of the project's stated objectives in its own right.

4.1 What the Models Project

Both fitted models agree on direction — continued, accelerating nominal growth through 2028-29 — but differ in magnitude because they are built differently. The exploratory time-only model projects Value of Output reaching approximately Rs. 1,883,921,611 lakh by 2028-29 ($R^2 = 0.974$ on history). The primary Model 1, which additionally conditions on projected Fixed Capital and Workers growth alongside time, projects a steeper Rs. 2,401,084,574 lakh by the same year ($R^2 = 0.995$ on history) — roughly 27% higher than the time-only projection, because it effectively assumes capital and labour continue expanding at their own recent trajectories, not just that 'time passes'. Read together, the two models bracket a plausible 2028-29 output range of roughly Rs. 1.88–2.40 crore lakh, implying nominal industrial output could be 25–35% higher in five years than the Rs. 1,532,716,610 lakh recorded in 2023-24.

4.2 Reading the Forecast Correctly

Three points matter for interpreting these numbers responsibly:

- Nominal, not real, growth. Neither model adjusts for inflation. Some portion of the projected increase reflects price growth rather than physical output growth; the true real-terms industrial expansion is smaller than the headline nominal figures suggest.
- Extrapolation through an unusual period. Both models are fit on a series that includes the 2020-21 COVID-19 contraction and an unusually steep 2021-22 to 2023-24 rebound. A polynomial fit extrapolating through that rebound will tend to project a continuation of exceptional recent growth rates rather than reverting to the more moderate pre-COVID trend — which is why both projections should be read as upper-range trend-continuation scenarios rather than central-case forecasts.
- No structural break is modeled. Neither model can anticipate policy shifts, global trade conditions, commodity price cycles, or technology shocks; they purely extrapolate the mathematical shape of the historical curve. The gap between the two models' 2028-29 estimates (roughly Rs. 52 crore lakh) is itself a useful, data-driven estimate of how sensitive the forecast is to modeling choices, and is a more honest way to communicate uncertainty than either point estimate alone.

4.3 Where the Growth Is Most Likely to Concentrate

Combining the national forecast with the structural findings elsewhere in this report suggests projected growth is unlikely to be evenly distributed. Given that Fixed Capital is the dominant driver of sector-level NVA (Section 3.5) and that capital has been growing faster than any other national input (Section 3.1), industries that are already capital-intensive and already large — Basic Metals, Motor Vehicles, Chemicals, and Pharmaceuticals — are the most probable near-term beneficiaries of continued capital-led growth, absent any deliberate policy effort to broaden participation. Similarly, because Workers is the dominant driver of state-level NVA and the three largest states (Maharashtra, Gujarat, Tamil Nadu) already combine industrial scale on every input dimension (Section 3.6), those states are structurally best positioned to capture a disproportionate share of aggregate growth unless smaller, currently under-performing or below-frontier states (Section 3.7) are specifically targeted for investment.

5. Key Findings

- Industrial output, NVA, and fixed capital grew at 10.51%, 9.72%, and 12.35% CAGR respectively from 1981-82 to 2023-24 (nominal), versus only 2.25% for employment — four decades of capital-led, not employment-led, industrial growth.
- The top 5 industries (Basic Metals, Motor Vehicles, Chemicals, Pharmaceuticals, Machinery) generate 43.9% of national industrial NVA; the top 5 states (Maharashtra, Gujarat, Tamil Nadu, Karnataka, Uttar Pradesh) generate 54.9%.
- Despite this concentration, both HHI measures (810 for states, 628 for sectors) classify India's industrial structure as unconcentrated/competitive, well below the 1,500 threshold.
- At the state level, Workers is the strongest predictor of NVA (Gradient Boosting, LOO-CV $R^2 = 0.930$); at the sector level, Fixed Capital dominates instead (Random Forest, LOO-CV $R^2 = 0.815$) — independently confirmed by SHAP attribution.
- Rajasthan, Jammu & Kashmir, Sikkim (states) and Pharmaceuticals, Basic Metals, Chemicals (sectors) are the clearest over-performers relative to model-predicted NVA; Puducherry, Punjab, Delhi (states) and Coke & Refined Petroleum, Textiles, Computer/Electronic/Optical Products (sectors) are the clearest under-performers.
- Coke and Refined Petroleum Products is a productivity outlier — Rs. 95.6 lakh NVA per worker at Rs. 471.5 lakh capital per worker, roughly 4-5× any other industry on both measures; Sikkim is the analogous state-level outlier at Rs. 73.4 lakh NVA per worker.
- K-Means clustering independently recovers a 'big-3' state cluster (Maharashtra, Gujarat, Tamil Nadu) and a 'big-4' sector cluster (Basic Metals, Chemicals, Food Products, Coke & Refined Petroleum) using only input-scale data — corroborating the descriptive concentration findings from a completely independent method.
- National output is projected to reach Rs. 1.88-2.40 crore lakh by 2028-29 under two related polynomial trend models ($R^2 = 0.974$ and 0.995 respectively on history), implying continued double-digit nominal growth, best read as an upper-range trend-continuation scenario given the models' extrapolation through the post-COVID rebound.

6. Limitations

- Very small cross-sectional samples. The state and sector ML models are trained on only 36 and 29 observations respectively. LOO-CV is the right tool for a sample this small, but R^2 values from 29-36 points still carry wide uncertainty, and the models should be read as informative rather than definitive — a handful of unusual states or sectors can meaningfully shift the fitted relationship.
- Nominal, not inflation-adjusted, figures throughout. Every trend, growth rate, and forecast in this report uses current-price Rs. Lakh values exactly as ASI publishes them. Part of the reported 10-12% CAGRs and the entire five-year forecast reflect price growth as well as real output growth; a real-terms (deflated) version of this analysis would show smaller growth rates and would be needed before drawing conclusions about physical industrial capacity.
- The national forecast extrapolates through an atypical period. Both time-trend models are fit on a series including the 2020-21 COVID-19 collapse and an unusually steep recovery; a polynomial curve fit through that period will tend to project continuation of exceptional recent growth. The forecast should be treated as an upper-range scenario, not a central-case prediction, as noted in Section 4.2.
- Table 5 (77 granular 3-digit NIC industries) was loaded and validated but not directly modeled. The cross-sectional ML models operate at the 29-industry major-group level; a finer-grained repetition of the same modeling pipeline at the 3-digit level would likely surface within-group heterogeneity (e.g. different sub-industries within 'Chemicals and Chemical Products') that this report cannot see.

- The efficiency frontier and K-Means clustering are both sensitive to the specific features used. Both were built on Fixed Capital, Workers, Total Input, and Number of Factories; including additional characteristics available in the source tables (e.g. gross value of plant & machinery, days worked, wages) could shift which units are flagged as frontier/efficient or how clusters are formed.
- Cross-sectional models identify association, not causation. Flagging a state or sector as an 'over-performer' means its NVA is higher than the model — trained only on Fixed Capital, Workers, Total Input, and Factories — would predict; it does not by itself identify why (skill levels, product mix, capacity utilization, energy costs, and policy environment are all plausible unmeasured contributors) and should be treated as a starting point for further investigation, not a completed diagnosis.
- No external macroeconomic context is incorporated. GDP, inflation, exchange rates, global commodity prices, and trade-policy variables are all outside the scope of this analysis, which is built entirely from ASI's own published tables; several of the more volatile-looking findings (e.g. the timing and depth of the COVID dip) would benefit from being read alongside those external series.

7. Conclusion

This project set out to assess India's long-run industrial trends, identify the structural drivers of industrial performance across states and sectors, and build predictive and explanatory models grounded entirely in ASI 2023-24 data — and each of the five stated objectives was addressed with concrete, quantified evidence. The trend analysis (Objective 1) established that four decades of industrial growth have been overwhelmingly capital-driven, with employment growth lagging output growth by roughly a factor of five. The structural composition analysis (Objective 2) identified the specific industries and states that dominate current production, while the HHI concentration measures showed that dominance, while real, does not amount to an uncompetitive market structure. The productivity and efficiency analysis (Objective 3) went a level deeper, identifying not just which units are largest but which are most efficient at converting capital and labour into value added, and quantifying the specific gap between frontier and non-frontier performers. The predictive and explanatory modeling (Objective 4) delivered both a five-year national output forecast and — arguably the report's most actionable finding — clear, SHAP-confirmed evidence that state-level and sector-level industrial growth are driven by genuinely different mechanisms (labour scale versus capital intensity), a distinction with direct implications for how state-focused and sector-focused industrial policy should be designed differently. Finally, the segmentation analysis (Objective 5) independently corroborated the report's concentration findings through unsupervised clustering and produced a specific, named list of over- and under-performing states and sectors as a starting point for targeted intervention.

Taken together, the analysis supports a clear policy narrative: India's industrial base is growing fast and increasingly capital-intensively, is meaningfully but not dangerously concentrated in a handful of large states and industries, and contains identifiable, evidence-based opportunities — both geographic and sectoral — where inputs are not yet converting into value added as efficiently as demonstrated peers show is achievable. The limitations in Section 6 — small sample sizes, nominal pricing, and the absence of granular 3-digit and macroeconomic context — mark out exactly what a fuller, more operationally deployable version of this study would need to address next.

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